

Mathematical Methods for International Commerce

Week 10/2: Unconstrained Optimization (5.4)

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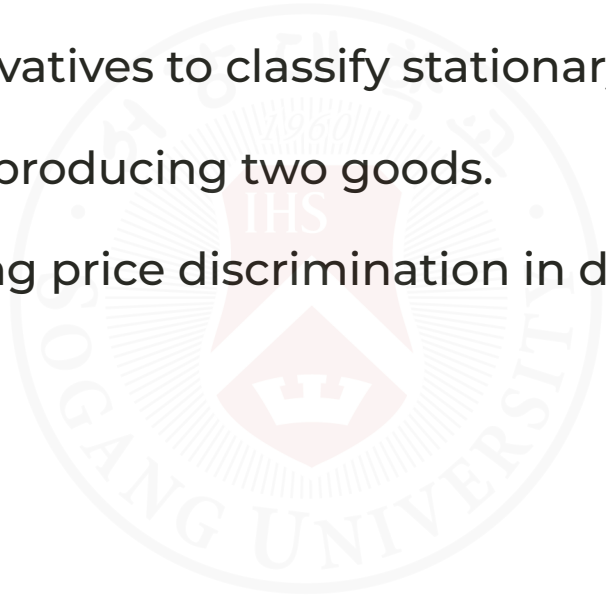
Why It Matters in Economics & Business

- Optimization helps firms maximize profit by adjusting inputs or pricing.
- It allows for strategic decision-making in pricing, production, and resource allocation.
- Applications in:
 - Cost minimization (minimizing production costs)
 - Profit maximization (maximizing revenue)
 - Price discrimination (optimizing prices in different markets)
 - Resource allocation (optimizing the use of limited resources)
 - Risk management (optimizing investment portfolios)

Learning Objectives

By the end of this class, you should be able to:

- Use first-order partial derivatives to find stationary points.
- Use second-order partial derivatives to classify stationary points.
- Maximize the profit of a firm producing two goods.
- Optimize profit for a firm using price discrimination in different markets.



Agenda

1. Unconstrained Optimization (5.4)
2. Class Activity



Finding Stationary Points

- Our first step is to find the stationary points of a function.
 - A stationary point is where the first derivative (or partial derivative) is zero.
- Why? - Because it indicates a potential maximum or minimum.
- We can find stationary points using the first-order conditions.

For a function of two variables $f(x, y)$, the first-order partial derivatives are:

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}$$

Stationary points occur where both derivatives equal zero:

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

Stationary points are candidates for extrema or saddle points.

Example:

Let $f(x, y) = 3x^2 - 2xy + y^2$

1. Calculate f_x and f_y .
2. Set $f_x = 0$ and $f_y = 0$ to find the stationary points.

Note: The function is a quadratic form, which often has a single stationary point that can be classified as a maximum, minimum, or saddle point based on the second-order conditions.

Solution: Stationary Points

1. First-order derivatives:

$$f_x = 6x - 2y, \quad f_y = -2x + 2y$$

Setting them to zero:

$$6x - 2y = 0 \quad -2x + 2y = 0$$

Solving simultaneously:

- From $6x - 2y = 0$, we get $y = 3x$.
- From $-2x + 2y = 0$, we get $y = x$.
- Equating them gives $3x = x$, so $x = 0$ and therefore $y = 0$.

Stationary point: $(x, y) = (0, 0)$.

It means that at this point, the slope of the function is zero in both directions.

- This is a candidate for a local maximum, minimum, or saddle point.

Second-Order Partial Derivatives and Hessian Matrix

To classify the stationary points, we use the second-order partial derivatives:

$$f_{xx}, f_{yy}, f_{xy}$$

The determinant of the Hessian matrix D is given by:

$$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

- If $D > 0$ and $f_{xx} > 0$: Local Min
- If $D > 0$ and $f_{xx} < 0$: Local Max
- If $D < 0$: Saddle Point
- If $D = 0$: Inconclusive

Note 1: The Second-order conditions provide sufficient conditions for local extrema when $D \neq 0$.

- They help us classify the nature of the stationary point.
- If $D = 0$, we cannot conclude anything about the nature of the stationary point.
- We may need to use higher-order derivatives or other methods to classify the point.

Note 2: The second-order conditions are based on the assumption that the function is twice differentiable.

- If the function is not twice differentiable, we may need to use other methods to classify the point.

Second-Order Partial Derivatives and Hessian Matrix (cont'd)

Hessian Matrix

What is the Hessian Matrix?

- The Hessian matrix, denoted by H_x , is a square matrix of second-order partial derivatives of a scalar-valued function $f(x_1, x_2)$.
- It provides information about the **curvature** of the function and helps in determining the nature of stationary points.
- The determinant of the Hessian matrix is used to classify stationary points.

Definition:

$$H_x = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

Explanation:

- **Diagonal Elements:** Measure the curvature with respect to each variable independently.
- **Off-Diagonal Elements:** Measure how the function curvature changes when both variables interact.

Second-Order Partial Derivatives and Hessian Matrix (cont'd)

Hessian Matrix

Why is it Important?

- The Hessian matrix helps to determine the nature of stationary points:
 - Positive determinant and positive diagonal elements: **Local minimum.**
 - Positive determinant and negative diagonal elements: **Local maximum.**
 - Negative determinant: **Saddle point.**
 - Zero determinant: **Inconclusive test.**

Example: Classification

Continuing from the previous example:

Second-order derivatives:

$$f_{xx} = 6, \quad f_{yy} = 2, \quad f_{xy} = -2$$

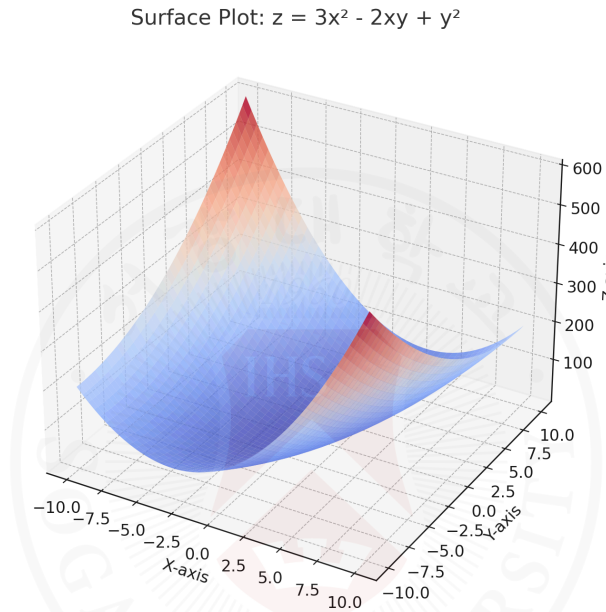
Determinant:

$$D = (6)(2) - (-2)^2 = 12 - 4 = 8$$

Since $D > 0$ and $f_{xx} > 0$, the point $(0, 0)$ is a **local minimum**. It means that the function has a minimum value at this point.

- The function is locally convex (positive definite Hessian).
- The function is increasing in both directions away from this point.
- The function is "bowl-shaped" at this point.
- The function has a unique minimum value at this point.

Visualizing the Function



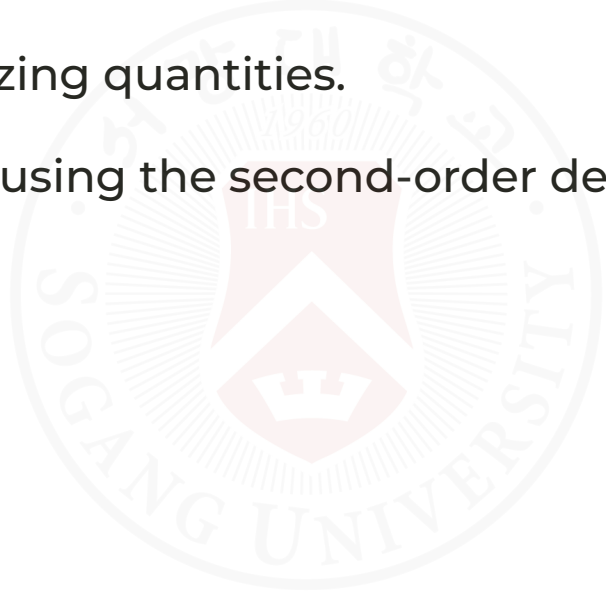
- The blue regions indicate lower values of z , while the red regions indicate higher values.
- The shape is not symmetric due to the interaction term $-2xy$, creating a slanted surface.
- The surface is bowl-shaped with a minimum at $(0,0)$, confirming our analytical result.

Economic Application (1): Profit Maximization

A firm produces two goods Q_1 and Q_2 with profit function:

$$\Pi(Q_1, Q_2) = 200Q_1 + 300Q_2 - 50Q_1^2 - 75Q_2^2 - 30Q_1Q_2$$

1. Determine the profit-maximizing quantities.
2. Classify the stationary points using the second-order derivatives.



Solution: Profit Maximization

1. First-order derivatives:

$$\Pi_{Q_1} = 200 - 100Q_1 - 30Q_2$$

$$\Pi_{Q_2} = 300 - 150Q_2 - 30Q_1$$

Setting to zero:

$$200 - 100Q_1 - 30Q_2 = 0$$

$$300 - 150Q_2 - 30Q_1 = 0$$

Solving simultaneously:

- $Q_1 \approx 1.49$
- $Q_2 \approx 1.70$

Solution: Profit Maximization (cont'd)

1. Second-order derivatives:

$$\Pi_{Q_1Q_1} = -100, \quad \Pi_{Q_2Q_2} = -150, \quad \Pi_{Q_1Q_2} = -30$$

Hessian Determinant:

$$D = (-100)(-150) - (-30)^2 = 15000 - 900 = 14100$$

Since $D > 0$ and $\Pi_{Q_1Q_1} < 0$, the point $(1.49, 1.70)$ is a **local maximum**.

It means that the firm maximizes its profit at this point.

The logo of Gangneung University is a circular seal. It features a central shield with a crown on top and the letters 'IHS' inside. The year '1960' is inscribed above the shield. The outer ring of the seal contains the university's name in Korean '강릉대학교' at the top and 'GANGNEUNG UNIVERSITY' at the bottom.

2. Group Activity

Your turn: Profit Maximization (Group 1)

A firm produces two goods Q_1 and Q_2 with the following profit function:

$$\Pi(Q_1, Q_2) = 150Q_1 + 250Q_2 - 40Q_1^2 - 60Q_2^2 - 20Q_1Q_2$$

Tasks:

1. Determine the profit-maximizing quantities.
2. Classify the stationary points using the second-order derivatives.
3. Interpret the economic meaning of the results.

Your turn: Cost Optimization (Group 2)

A firm has the following cost function:

$$C(L, K) = 5L^2 + 4K^2 - 8L - 16K + 50$$

Tasks:

1. Find the cost-minimizing combination of labor L and capital K .
2. Use the Hessian matrix to classify the stationary point.
3. Provide an economic interpretation of the result.

Your turn: Risk Management: Optimizing Investment Portfolios (Group 3)

A firm is investing in two assets, A and B, with the following return function:

$$R = 0.08x + 0.12y - 0.5(0.06x^2 + 0.09y^2) - 0.04xy$$

where:

- x = Investment in Asset A
- y = Investment in Asset B

1. Determine the optimal allocation between assets A and B to maximize returns.
2. Verify the optimal solution using the second-order conditions.
3. Interpret the economic meaning of the results.

Your turn: Price Discrimination: Maximizing Profit in Multiple Markets (Group 4)

A monopolist sells a product in two markets, A and B. The revenue functions for each market are:

$$R_A = 200Q_A - 5Q_A^2$$

$$R_B = 300Q_B - 10Q_B^2$$

The cost function for the firm is given by:

$$C = 50 + 20(Q_A + Q_B)$$

1. Determine the optimal quantities to sell in each market to maximize profit.
2. Verify the optimal solution using the second-order conditions.
3. Interpret the economic meaning of the results in the context of price discrimination.

Your turn: Two-Variable Cost Minimization (Group 5)

A firm has cost function:

$$C(x, y) = x^2 + y^2 - 4x - 6y + 20$$

Tasks:

1. Find the stationary point.
2. Classify the stationary point using the Hessian.
3. Interpret the result economically.

Your turn: Two-Variable Profit Maximization (Group 6)

A firm has profit function:

$$\Pi(x, y) = 80x + 42y - 4x^2 - 2xy - 3y^2 - 100$$

Tasks:

1. Find the stationary point (x^*, y^*) .
2. Classify the stationary point using the Hessian.
3. Interpret the result economically.

Summary

- **Objective:** Maximize or minimize a function without constraints.
- **First-Order Derivative:** Used to find stationary points (set the derivative to zero).
- **Second-Order Derivative:** Determines the nature of the stationary points:
 - $f''(x) > 0$: Local minimum
 - $f''(x) < 0$: Local maximum
- **Applications in Economics:**
 - Cost minimization (input allocation)
 - Profit maximization (optimal production)
 - Risk management (investment allocation)
 - Price discrimination (market segmentation)
- **Key Takeaways:**
 - Identify objective functions and derive first and second derivatives.
 - Analyze the Hessian matrix for functions of two variables.
 - Apply economic interpretations to the mathematical solutions.

Next Classes

- (May 12) Constrained Optimization (5.5)





Any QUESTIONS?

Thank you for your attention and participation!